Cloud Computing Infrastructure: Kubernetes Microservices, Serverless Deployment, and Cyber Security

1. Abstract & Introduction

Modern Cloud Computing infrastructure relies on microservices architecture, Docker containers, and Kubernetes orchestration. Securing cloud environments requires Zero Trust Architecture, Public Key Infrastructure (PKI), AES-256 encryption, and identity management.

2. Cloud Infrastructure & Security Protocols

Infrastructure as Code (IaC) templates define Virtual Private Clouds (VPC), auto-scaling groups, and API Gateways. Intrusion Detection Systems (IDS) monitor network traffic while SIEM logging aggregates security event telemetry across distributed clusters.

3. Resilience & Security Analysis

Chaos engineering tests validated 99.99% service availability during multi-region failover scenarios. Penetration testing confirmed zero unauthorized access vulnerabilities under simulated DoS attacks.

4. Conclusion

Combining container orchestration with automated threat detection ensures scalable, resilient, and secure enterprise cloud deployments.